Psychological statistics

Homework Assignment ④

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WII

1. Hypothesis

- **Comparison between Athletes and Non-Athletes:**

  H0: There is no difference in pain scores between athletes and non-athletes.

  Ha: There is a difference in pain scores between athletes and non-athletes.

- **Comparison between Stretching and No Stretching:**

  H0: There is no difference in pain scores between those who stretched and those who did not stretch.

  Ha: There is a difference in pain scores between those who stretched and those who did not stretch.

2. Check Normality and Homogeneity

- **Normality**

```
# A tibble: 4 × 5
  athlete     stretch       variable statistic       p
  <chr>       <chr>         <chr>        <dbl>   <dbl>
1 Athlete     No Stretching injury       0.911 0.0157
2 Athlete     Stretching    injury       0.888 0.00441
3 Non-Athlete No Stretching injury       0.949 0.162
4 Non-Athlete Stretching    injury       0.939 0.0866
```

The results indicate that the distribution of injury scores does not follow a normal distribution in “Athlete” groups. ($p < 0.05$)

- **Homogeneity of Variance**

```
Levene's Test for Homogeneity of Variance (center = median)
       Df F value    Pr(>F)
group   3  14.178 6.199e-08 ***
      116
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

The results indicate that the variances are not equal among different groups ($p < 0.05$)

3. Kruskal-Wallis test and Post-hoc Tests

Comparison between athletes and non-athletes:

```
        Kruskal-Wallis rank sum test

data:  injury by athlete
Kruskal-Wallis chi-squared = 23.985, df = 1, p-value = 9.709e-07
```

```
# A tibble: 1 × 7
  .y.    group1  group2         n1    n2 statistic           p
* <chr>  <chr>   <chr>       <int> <int>     <dbl>       <dbl>
1 injury Athlete Non-Athlete    60    60      886. 0.000000984
```

Kruskal-Wallis Test Results:

Since the p-value is very small (much less than 0.05), we reject the null hypothesis, indicating a significant difference in injury severity between athletes and non-athletes.

Wilcoxon Rank-Sum Test Results:

Since the p-value is very small (much less than 0.05), we again reject the null hypothesis, further confirming that there is a significant difference in injury severity between athletes and non-athletes.

Comparison between the prevention program group and the control group:

```
        Kruskal-Wallis rank sum test

data:  injury by stretch
Kruskal-Wallis chi-squared = 1.6935, df = 1, p-value = 0.1931
```

```
# A tibble: 1 × 7
  .y.    group1        group2        n1    n2 statistic     p
* <chr>  <chr>         <chr>      <int> <int>     <dbl> <dbl>
1 injury No Stretching Stretching    60    60      2043 0.194
```

Kruskal-Wallis Test Results:

Since the p-value is greater than 0.05, we fail to reject the null hypothesis, indicating no significant difference in injury severity between the stretching and no stretching groups.

4. Effect Size

```
> data %>% kruskal_effsize(injury ~ athlete)
# A tibble: 1 × 5
  .y.        n effsize method  magnitude
* <chr>  <int>   <dbl> <chr>   <ord>
1 injury   120   0.195 eta2[H] large
> data %>% kruskal_effsize(injury ~ stretch)
# A tibble: 1 × 5
  .y.        n effsize method  magnitude
* <chr>  <int>   <dbl> <chr>   <ord>
1 injury   120 0.00588 eta2[H] small
```

5. Conclusion

In summary, there is a significant difference in injury severity between athletes and non-athletes, while there is no significant difference in injury severity between the prevention program group and the control group.

MUSICAL TASTES

1. Hypothesis

H0-1: Age has no effect on music liking, the mean music ratings of the young group and the older group are equal.

H0-2: Music genre has no effect on music liking, the mean music ratings of the three music genres are equal.

H0-3: There is no interaction between age and music genre, the effect of age on music ratings does not vary by music genre.

2. Check Normality and Homogeneity

- **Normality**

```
# A tibble: 6 × 5
  music       age   variable statistic      p
  <chr>       <chr> <chr>        <dbl>  <dbl>
1 ABBA        old   liking       0.954 0.593
2 ABBA        young liking       0.963 0.751
3 Barf Grooks old   liking       0.954 0.591
4 Barf Grooks young liking       0.981 0.975
5 Fugazi      old   liking       0.915 0.162
6 Fugazi      young liking       0.886 0.0588
>
```

The p-values of the Shapiro-Wilk test are all greater than 0.05, except for the "Fugazi - young" group, where the p-value is 0.0588. Although this is close

to 0.05, it is still greater than 0.05. This indicates that the music likiing for most groups follow a normal distribution.

- **Homogeneity of Variance**

```
# A tibble: 2 × 5
  age     df1   df2 statistic     p
  <chr> <int> <int>     <dbl> <dbl>
1 old       2    42     0.859 0.431
2 young     2    42     1.53  0.228
>
```

The p-values of the Levene's test are all greater than 0.05, indicating that the music ratings have equal variances across different age groups.

Since most groups follow a normal distribution and all groups have equal variances, we can use a **two-way ANOVA** to examine the effects of age and music genre on music ratings, as well as the interaction between them.

3. Two-way ANOVA and Post-hoc Tests

```
ANOVA Table (type II tests)

     Effect        SSn      SSd DFn DFd       F        p p<.05      ges
1       age      0.711 32553.47   1  84   0.002 9.66e-01       2.18e-05
2     music  81864.067 32553.47   2  84 105.620 1.18e-23     * 7.15e-01
3 age:music 310790.156 32553.47   2  84 400.977 1.07e-43     * 9.05e-01
```

- The effect of age was not significant, F (1, 84) = 0.002, p = 0.966, indicating that age does **not have a significant influence** on music liking.
- The effect of music type was significant, F (2, 84) = 105.620, p < 0.001, suggesting that different music types **have a significant impact** on music liking.
- The interaction effect between age and music type was also significant, F (2, 84) = 400.977, p < 0.001, indicating that the combination of age and music type significantly influences music liking.

Post-hoc Tests

1. Young Group

```
# A tibble: 3 × 9
  .y.    group1      group2         n1    n2        p p.signif    p.adj p.adj.signif
* <chr>  <chr>       <chr>       <int> <int>    <dbl> <chr>       <dbl> <chr>
1 liking ABBA        Barf Grooks    15    15 4.69e- 2 *        1.41e- 1 ns
2 liking ABBA        Fugazi         15    15 1.41e-22 ****     4.23e-22 ****
3 liking Barf Grooks Fugazi         15    15 3.04e-24 ****     9.12e-24 ****
```

- ABBA vs. Barf Grooks: $p < 0.05$, significant difference
- ABBA vs. Fugazi: $p < 0.05$, significant difference
- Barf Grooks vs. Fugazi: $p < 0.05$, significant difference.

2. Old Group

```
# A tibble: 3 × 9
  .y.    group1      group2         n1    n2        p p.signif    p.adj p.adj.signif
* <chr>  <chr>       <chr>       <int> <int>    <dbl> <chr>       <dbl> <chr>
1 liking ABBA        Barf Grooks    15    15 1.03e-21 ****     3.09e-21 ****
2 liking ABBA        Fugazi         15    15 7.81e- 1 ns       1   e+ 0 ns
3 liking Barf Grooks Fugazi         15    15 5.84e-22 ****     1.75e-21 ****
```

- ABBA vs. Barf Grooks: $p < 0.05$, significant difference
- ABBA vs. Fugazi: $p > 0.05$, no significant difference
- Barf Grooks vs. Fugazi: $p < 0.05$, significant difference.

4. Effect Size

```
> omega_factorial(45,2,3,0.711,81864.067,310790.156,32553.47)
[1] "Omega-Squared A:  -0.000288239966250956"
[1] "Omega-Squared B:  0.191891287165971"
[1] "Omega-Squared AB:  0.730120820067289"
```

5. Conclusion

The two-way analysis of variance (ANOVA) indicates significant effects of age, music type, and their interaction on music ratings. Further post-hoc tests reveal significant differences among different music types, and these differences may vary between the young and old groups. Specifically, in the young group, differences among all music types are significant, whereas in the old group, differences between ABBA and Barf Grooks, and between Barf Grooks and Fugazi are significant, but the difference between ABBA and Fugazi is not. The Omega-squared value for the interaction between age and music type (AB) is 0.730, indicating that this interaction accounts for 73.0% of the total variance, which demonstrates that the **interaction effect is quite significant**.

TEXT MESSAGE

In this scenario, we can use **two-way repeated measures ANOVA.** This is because there are two independent variables (factors) being manipulated:

1. Text messaging use (encouraged vs. forbidden)
2. Time point (before vs. after the experiment)

The dependent variable is the grammatical test scores measured before and after the experiment. By using a two-way repeated measures ANOVA, we can assess the main effects of text messaging use and time point on the grammatical test scores, as well as any interaction effect between these two factors.

1. Hypothesis

- H01: There is no significant difference in mean scores among different groups across different time points.
- H02: There is no significant difference in mean scores across different time points within each group.
- H03: There is no significant difference in the change of mean scores across different time points among different groups.

2. Check Normality

Three groups show p-values greater than 0.05, but for the "Text Messagers" group at "Six_months," the p-value for the score is less than 0.05, indicating non-normal distribution. As most of the data conforms to a normal distribution, a 2-way Repeated-measures ANOVA will be used for further analysis.

```
# A tibble: 4 × 5
  Group          Time       variable statistic       p
  <chr>          <chr>      <chr>        <dbl>   <dbl>
1 Controls       Baseline   Score        0.971 0.669
2 Controls       Six_months Score        0.942 0.161
3 Text Messagers Baseline   Score        0.962 0.448
4 Text Messagers Six_months Score        0.877 0.00606
```

3. Two-way repeated measures ANOVA.

```
Error: Subject
          Df Sum Sq Mean Sq F value Pr(>F)
Group      1    581   580.8   2.987 0.0904 .
Residuals 48   9334   194.5
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Error: Subject:Time
           Df Sum Sq Mean Sq F value   Pr(>F)
Time        1   1529  1528.8  15.457 0.000271 ***
Group:Time  1    412   412.1   4.166 0.046752 *
Residuals  48   4748    98.9
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- Error term of Subject:

  Group: The F-value for the factor Group is 2.987, with a p-value of 0.0904, indicating non-significance ($p > 0.05$). This suggests that the variance difference between different groups is **not significant.**

- Error term of Subject:Time:

  Time: The F-value for the factor Time is 15.457, with a p-value of 0.000271, indicating high significance ($p < 0.001$). This implies that the variance differs significantly across different time points.

  Group:Time: The F-value for the interaction between Group and Time is 4.166, with a p-value of 0.046752, indicating significance ($p < 0.05$). This suggests that the interaction between group and time **significantly affects the results.**

4. Post-hoc Tests

```
# A tibble: 2 × 11
  Group          .y.   group1   group2    n1    n2 statistic    df     p p.adj p.adj.signif
* <chr>          <chr> <chr>    <chr>  <int> <int>     <dbl> <dbl> <dbl> <dbl> <chr>
1 Controls       Score Baseline Six_m…    25    25      2.01    24 0.056 0.056 ns
2 Text Messagers Score Baseline Six_m…    25    25      3.38    24 0.002 0.002 **
```

The results indicate that, in the Controls group, the difference in scores between Baseline and Six_months is not statistically significant ($p = 0.056$). However, in the Text Messagers group, the difference in scores between Baseline and Six_months is significant ($p = 0.002$).

5. Conclusion

In the “Text Messagers” group, there was a significant regress in English grammar scores after six months, while in the Controls group, this improvement was not as pronounced. This may imply a positive impact of sending text messages on children's learning of English grammar.